

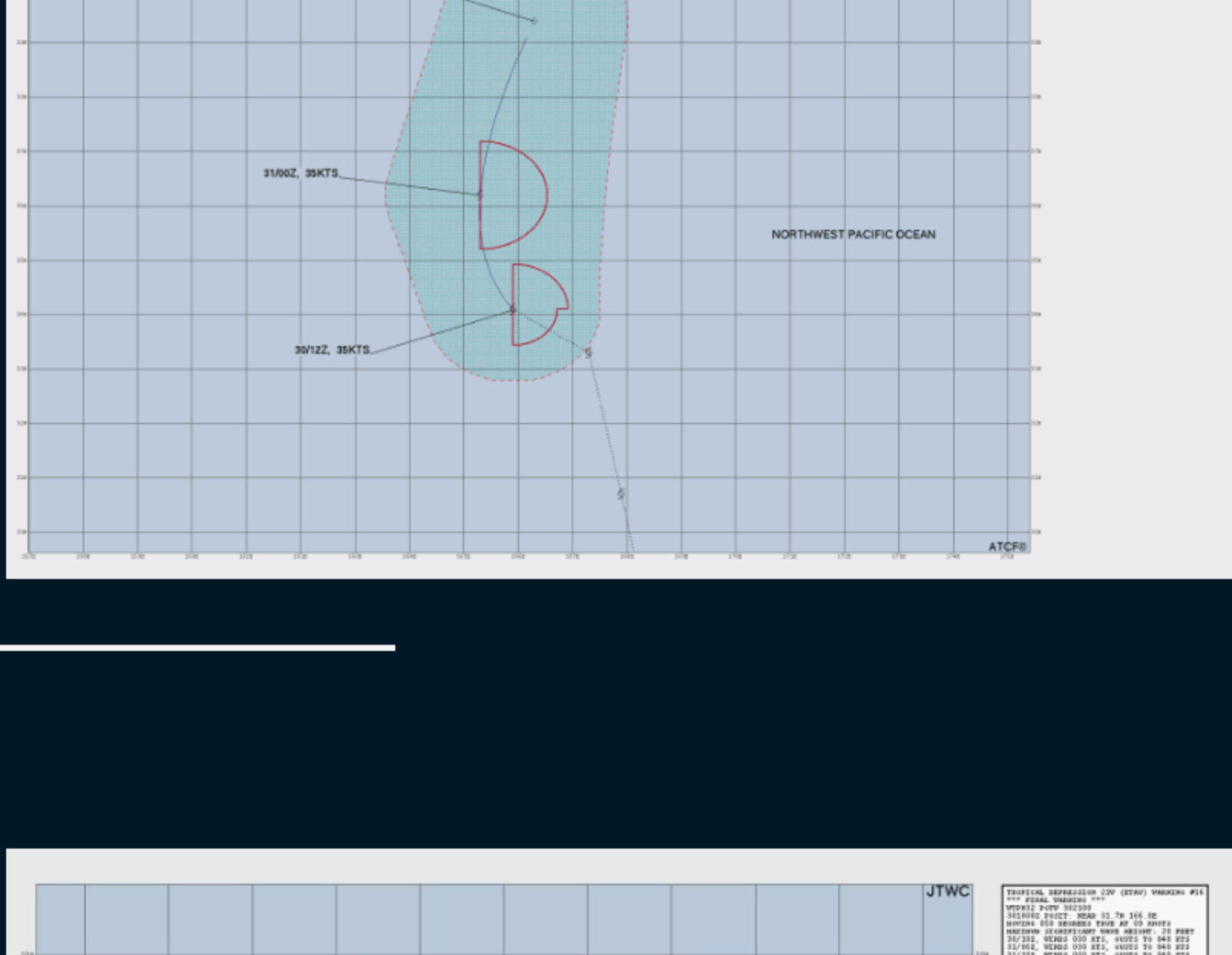
Home ▾Indian Ocean ▾West Pacific ▾South Pacific ▾Central Pacific ▾East Pacific ▾Atlantic ▾

Tropical Storm BANG-LANG

1. TROPICAL STORM 20W (BANG-LANG) WARNING NR 018
82 ACTIVE TROPICAL CYCLONES IN NORTHWESTPAC
MAX SUSTAINED WINDS BASED ON ONE-MINUTE AVERAGE
WIND RADII VALID OVER OPEN WATER ONLY

WARNING POSITION:
301200Z --- NEAR 34.1N 165.9E
MOVEMENT PAST SIX HOURS - 305 DEGREES AT 14 KTS
POSITION ACCURATE TO WITHIN 600 NM
POSITION BASED ON CENTER LOCATED BY SATELLITE
PRESENT WIND DISTRIBUTION:
MAX SUSTAINED WINDS - 035 KT, GUSTS 045 KT
WIND RADII VALID OVER OPEN WATER ONLY
BECOMING EXTRATROPICAL
RADIUS OF 034 KT WINDS - 050 NM NORTHEAST QUADRANT
040 NM SOUTHEAST QUADRANT
080 NM SOUTHWEST QUADRANT
090 NM NORTHWEST QUADRANT

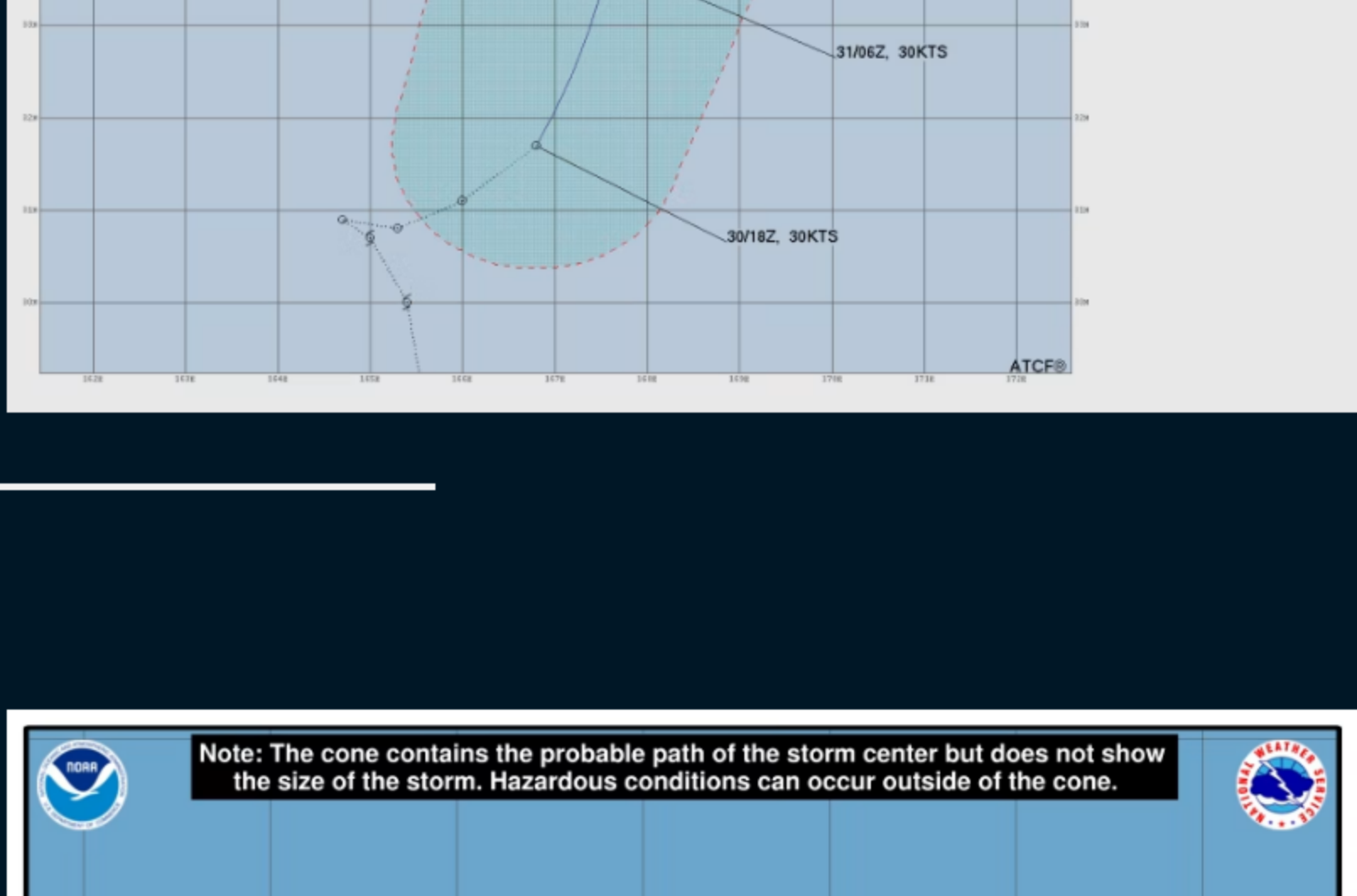
REPEAT POSIT: 34.1N 165.9E



Tropical Storm ETAU

1. TROPICAL DEPRESSION 21W (ETAU) WARNING NR 010
82 ACTIVE TROPICAL CYCLONES IN NORTHWESTPAC
MAX SUSTAINED WINDS BASED ON ONE-MINUTE AVERAGE
WIND RADII VALID OVER OPEN WATER ONLY

WARNING POSITION:
301800Z --- NEAR 31.7N 166.8E
MOVEMENT PAST SIX HOURS - 050 DEGREES AT 09 KTS
POSITION ACCURATE TO WITHIN 600 NM
POSITION BASED ON CENTER LOCATED BY SATELLITE
PRESENT WIND DISTRIBUTION:
MAX SUSTAINED WINDS - 030 KT, GUSTS 040 KT
WIND RADII VALID OVER OPEN WATER ONLY
BECOMING SUBTROPICAL
REPEAT POSIT: 31.7N 166.8E



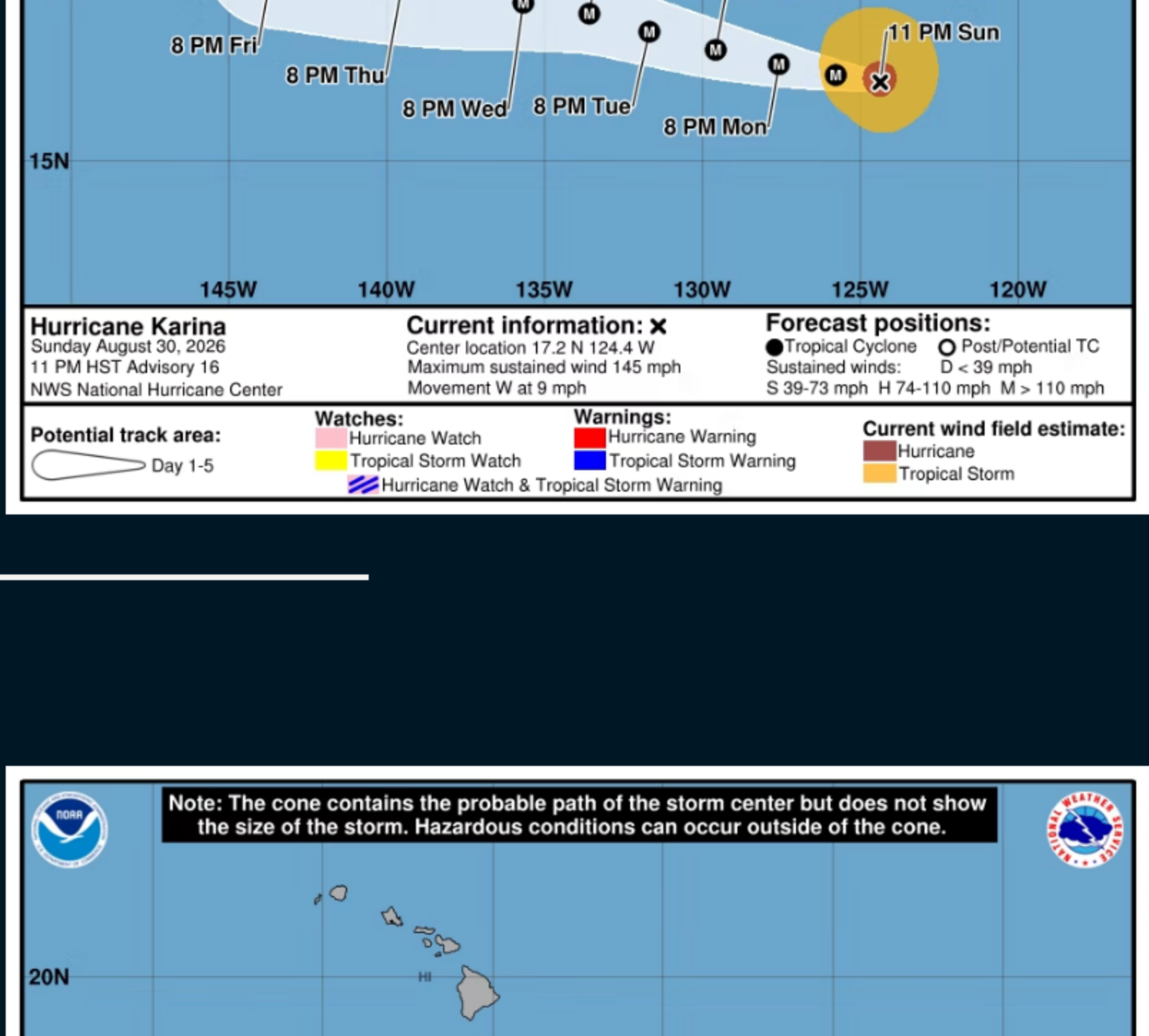
Hurricane KARINA

Hurricane Karina Advisory Number 16
NWS National Hurricane Center Miami FL
1100 PM HST Sun Aug 30 2026

...KARINA INTENSIFIES INTO A CATEGORY 4 HURRICANE AS IT WESTWARD...

SUMMARY OF 1100 PM HST...0900 UTC...INFORMATION

LOCATION...47.2N 124.4W
ABOUT 1020 MI...1640 KM WSW OF THE SOUTHERN TIP OF BAJA
MAXIMUM SUSTAINED WINDS...145 MPH...230 KM/H
PRESENT MOVEMENT...W OR 275 DEGREES AT 9 MPH...15 KM/H
MINIMUM CENTRAL PRESSURE...942 MB...27.82 INCHES



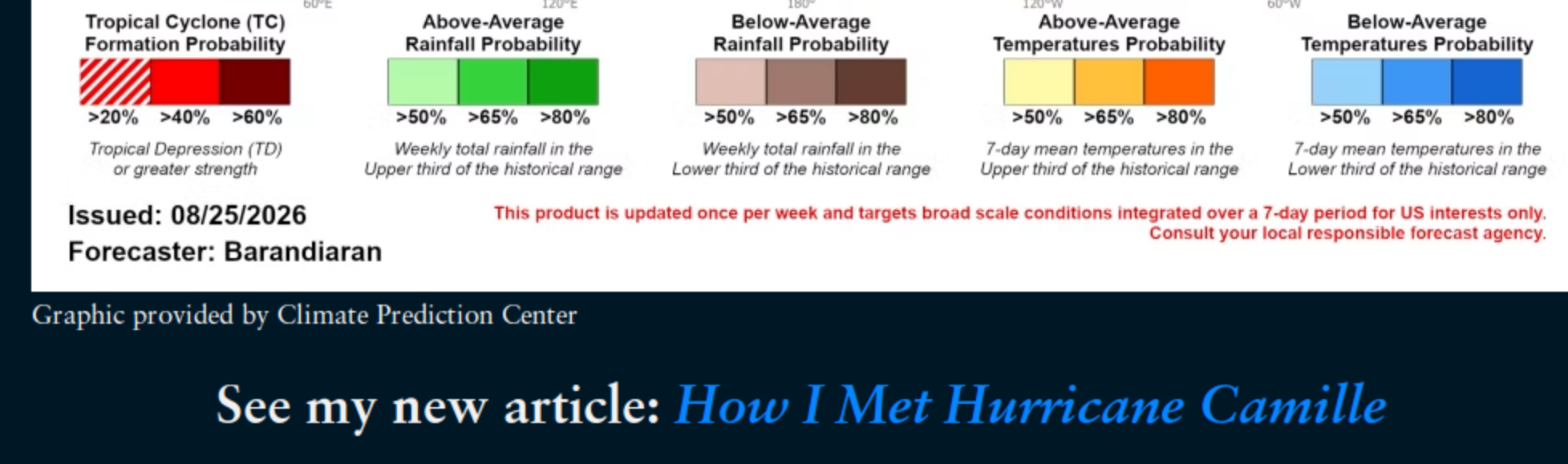
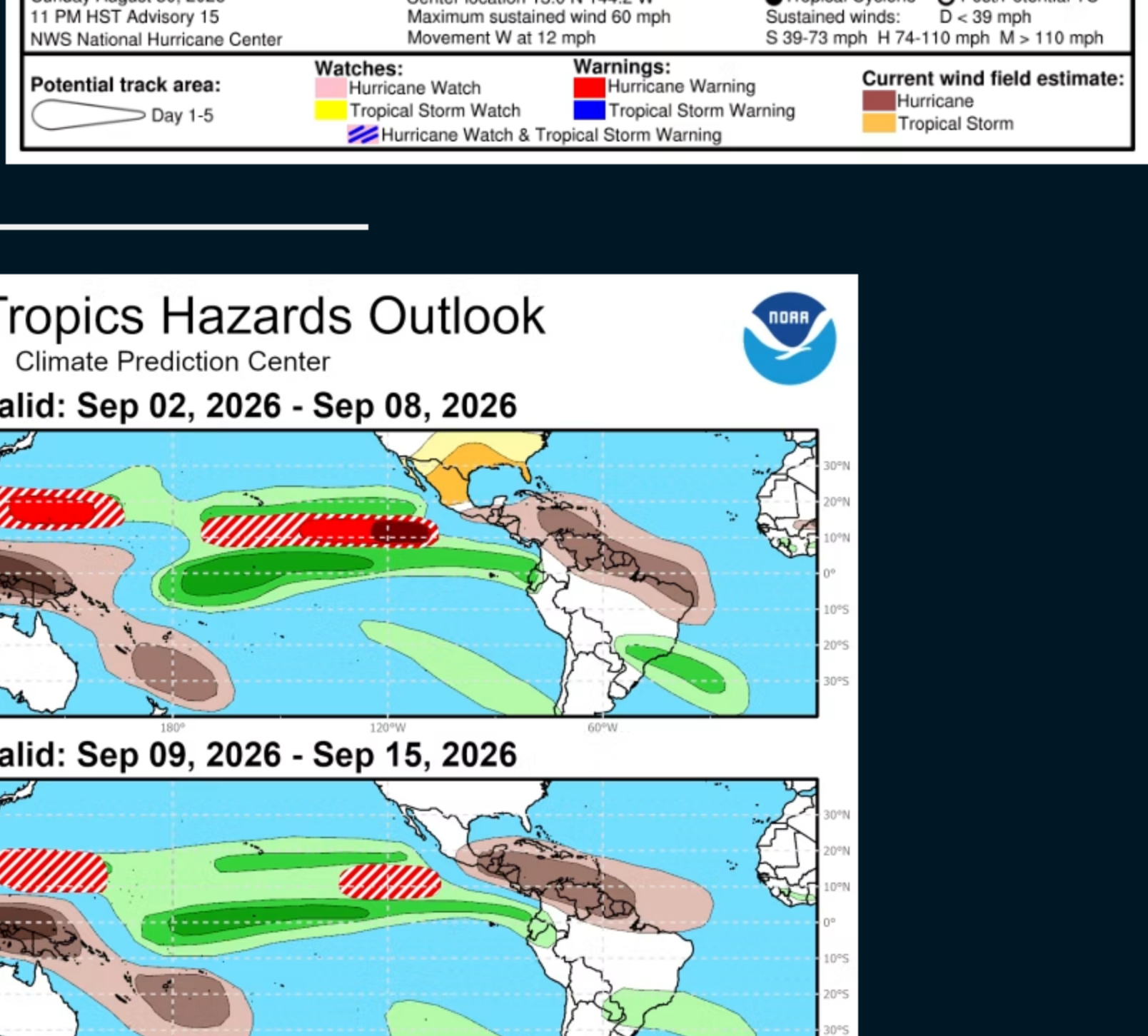
Tropical Storm LOWELL

Tropical Storm Lowell Advisory Number: 15
NWS Central Pacific Hurricane Center Honolulu HI
1100 PM HST Sun Aug 30 2026

...LOWELL STRENGTHENS SLIGHTLY ON A WESTWARD TRACK...

SUMMARY OF 1100 PM HST...0900 UTC...INFORMATION

LOCATION...13.0N 144.2W
ABOUT 855 MI...1380 KM ESE OF HILO HAWAII
MAXIMUM SUSTAINED WINDS...60 MPH...95 KM/H
PRESENT MOVEMENT...W OR 270 DEGREES AT 12 MPH...19 KM/H
MINIMUM CENTRAL PRESSURE...997 MB...29.44 INCHES



See my new article: [How I Met Hurricane Camille](#)

What Is a Hurricane?

A hurricane (or typhoon, or severe tropical cyclone), the strongest storm on Earth, is a cyclonic (rotary) storm that derives its energy from cloud formation and rainfall, unlike frontal cyclones that derive their power from a temperature gradient.

A hurricane begins as a tropical depression with a sustained wind speed of less than 39 mph (35 knots; 63 km/hr). As the system strengthens, it becomes a tropical storm with winds from 39 to 73 mph (35-63 knots; 63-118 km/hr). Tropical storms are named in the Atlantic, East, Central and Northwest Pacific, in the South Indian Ocean, and in the Arabian Sea. When the winds are sustained (based on a one-minute average) at 74 mph (64 knots; 119 km/hr), the storm becomes: In the Atlantic Ocean, East Pacific, Central Pacific (east of the International Dateline) and Southeast Pacific (east of 160°E) a Hurricane; in the Northwest Pacific (west of the International Dateline) a Typhoon; in the Southwest Pacific (west of 160°E) and Southeast Indian Ocean (east of 90°E) a Severe Tropical Cyclone; in the North Indian Ocean a Severe Cyclonic Storm; and in the Southwest Indian Ocean (west of 90°E) a Tropical Cyclone.

The Saffir-Simpson Hurricane Scale

Category 1 – 64-82 knots (74-95 mph; 119-153 km/h). Damage is limited to foliage, signage, unanchored boats and mobile homes. There is no significant damage to buildings. The main threat to life and property may be flooding from heavy rains.

Category 2 – 83-95 knots (96-110 mph; 154-177 km/h). Roof damage to buildings. Doors and windows damaged. Mobile homes severely damaged. Piers damaged by storm surge. Some trees blown down, more extensive limb damage.

Category 3 – 96-112 knots (111-129 mph; 178-208 km/h). Major Hurricane. Structural damage to some buildings. Mobile homes are completely destroyed. Roof damage is common. Storm surge begins to cause significant damage in beaches and harbors, with small buildings destroyed.

Category 4 – 113-136 knots (130-156 mph; 209-251 km/h). Structural failure of some buildings. Complete roof failures on many buildings. Extreme storm surge damage and flooding. Severe coastal erosion, with permanent changes to the coastal landscape not unheard of. Hurricane force winds extend well inland.

Category 5 – 137+ knots (157+ mph; 252+ km/h). Complete roof failure on most buildings. Many buildings destroyed, or structurally damaged beyond repair. Catastrophic storm surge damage. In the Northwest Pacific, a typhoon that reaches 150 mph (241 km/hr) is called a Super Typhoon.

SAFFIR-SIMPSON SCALE				
Category	Knots	MPH	KM/H	Damage
1	64-82	74-95	119-153	Minimal
2	83-95	96-110	154-177	Moderate
3	96-112	111-129	178-208	Extensive
4	113-136	130-156	209-251	Extreme
Super Typhoon	130+	150+	241+	Catastrophic
5	137+	157+	252+	Catastrophic

Storm Surge

Historically, storm surge is the primary killer in hurricanes. The exact storm surge in any given area will be determined by how quickly the water depth increases offshore. In deep-water environments, such as the Hawaiian islands, storm surge will be enhanced by the rapidly decreasing ocean depth as the wind-driven surge approaches the coast. The peak storm surge is on the right-front quadrant (left-front in the Southern Hemisphere) of the eyewall at landfall, where on-shore winds are the strongest, and at the leading edge of the eyewall. Contrary to a popular myth, the storm surge is entirely wind-driven water—it is not caused by the low pressure of the eye. Another factor in the severity of the storm surge is tide. Obviously, an 18-foot storm surge at high tide is that much worse than an 18-foot surge at low tide.

Tropical Cyclone Formation

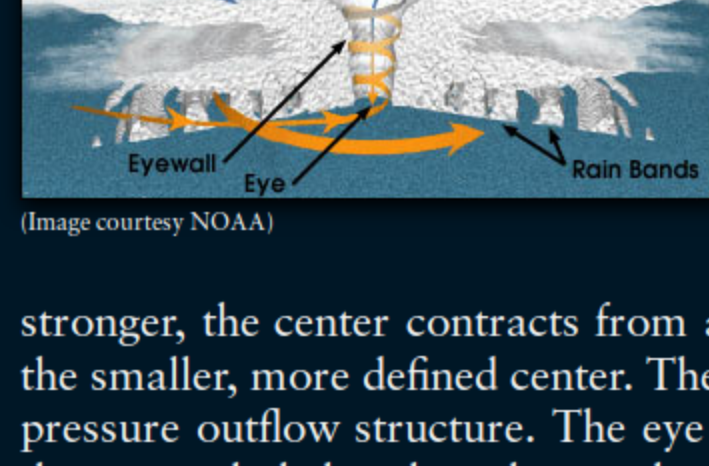
By Jonathan Edwards

Tropical Cyclone Genesis is the technical term for the process of storm formation that leads ultimately to what are called hurricanes, typhoons, or tropical cyclones in various parts of the world.

This occurs when, in the Northern Hemisphere, the Intertropical Convergence Zone, or ITCZ, shifts northward out of the doldrums and atmospheric conditions become favorable for tropical cyclone formation after about the middle of May.

A series of low-pressure ripples develops within the ITCZ. These are known as tropical waves and progress from east to west. In the late season, they typically shift their movement toward the west-northwest, or even northwest, after crossing 45° or 50° W longitude.

These tropical waves, ideally embedded in the deep-layer easterly flow, contain a northeast wind shift. This is typically referred to as a “convergence”, where lines of equal atmospheric pressure are pressed together between the high-pressure ridge to the north and the developing low-pressure system. The divergence that results ahead of the convergence zone gives us a northeasterly wind as the axis of the tropical wave approaches. Gusts up to 25 mph may occur. Sometimes there can be gusts to tropical storm force in stronger waves. There can be next to no weather associated with these waves, and they may pass virtually unnoticed. More typically, there are bands of disturbed weather riding the axis of the wave.



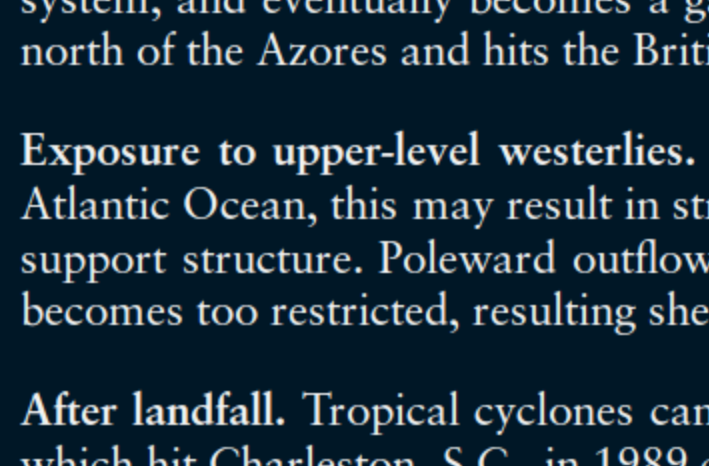
(Graphic by Robert Simmon, NASA GSFC)

When the wave passes over warmer waters (SSTs), convection and resulting rainfall are enhanced. This greater rainfall is concomitant with falling surface pressures. By the time these pressures fall to 1008mb, it is likely that the northeast wind has closed off to a southwest wind on the backside of the wave. The forward motion of the wave completes the closure on the northern side of a broad low-level center, and a tropical depression has formed.

We often hear that a tropical depression has formed, but conditions are unfavorable for further development. There are two conditions that must be present for the tropical depression to continue its development: warm SSTs (above 79° Fahrenheit/26° Celsius) and low vertical shear. A tropical cyclone derives its power from the warm waters below. In addition, a strong anticyclone directly above the low-level inflow is favorable. As a tropical cyclone is pulling in warm, moist air at the surface, it must also evacuate this inbound flow aloft. This occurs in the upper levels of the atmosphere, where high pressure facilitates the development of the cyclone by evacuating the flow from the lower levels of the cyclone. Every powerful hurricane has an equally powerful high pressure system over it. At the surface, the air spirals inward in a counter-clockwise direction, rises in the developing center, and spirals out at the top in a clockwise direction. In the Southern Hemisphere, it is reversed: clockwise inbound, counterclockwise outbound.

If the upper-level high pressure system does not remain over the cyclone, it means there is shear ahead. This is a strong jet of air that is blowing directly over the cyclone, ripping the cloud tops off the deep convection. This has the effect of breaking down the whole mechanism. This is known as vertical shear. Vertical shear usually comes from a westerly or southerly direction, and can occur if the cyclone is located in an unfavorable position near a cold front or upper-level low pressure system. Another factor that can interfere with the development of a tropical cyclone is subsidence. Subsidence is the sinking of air. This happens on the edge of an upper-level high-pressure system. Subsiding air has the effect of suppressing thunderstorm formation. This also is why a tropical storm that tries to form near an established hurricane has a very difficult time—the cyclone is on the edge of the hurricane's upper-level outflow, and may have to contend with both subsidence and shear. The effect of shear on a cyclone can range from a failure to develop further to catastrophic collapse of the tropical cyclone's support structure.

Tropical cyclones have the low-level circulation and the upper-level circulation (outflow), whose formation was discussed above. There is also a mid-level circulation. The mid-level circulation is similar in structure to the low-level circulation, and is critical to the survival of a tropical cyclone that is passing over land. The low-level circulation can be severely disrupted, or even dissipated, by interaction with land, especially mountainous terrain. If the mid-level circulation remains intact, the cyclone can regenerate rapidly when it reemerges over water, providing other factors are favorable.



(Image courtesy NOAA)

Assuming all the ingredients are in place—warm SSTs, upper-level high pressure, and falling surface pressures—the cyclone will develop and reach a point of rapid intensification. It is one of nature's perfect machines. As warm waters feed the convection swirling around the center, heavy rainfall lowers surface pressures, high pressure aloft evacuates the inflow, which intensifies the inflow of warm, moist air, which in turn increases the rainfall and brings about a more rapid fall in central pressure.

Eye formation begins when a tropical storm reaches approximately 65mph, provided conditions are favorable for strengthening to continue. The eyewall begins to make its appearance, usually on the eastern (Northern Hemisphere) edge of the center. As the system becomes better organized and stronger, the center contracts from about 200 miles across to roughly 90 miles at this stage. An increase in rotational velocity accompanies the smaller, more defined center. The inflow is spiralling in ever faster as it is evacuated up through the developing eyewall and out by the high pressure outflow structure. The eye begins to appear as a clear spot in the center, as the air here is sinking. The eyewall creates subsidence that not only helps clear the eye, but can also produce a feature known as the “moat”, which is an area of relatively weak convection outside the core of the tropical cyclone. The combination of eye and moat can make the storm's core look like a doughnut. The eye can have a diameter anywhere from 10 to 40 miles across.